NOVA IMS INFORMATION MANAGEMENT SCHOOL

# F1 Race Strategy Intelligence Platform

PySpark Big Data Platform · Formula 1 Constructor Decision-Making

NOVA IMS · Big Data Analytics 2026

5th June 2026

## THE CLIENT CHALLENGE

*Why Formula 1 is a Big Data Problem. 74 years of race data, millisecond decisions*

Formula 1 generates one of the densest operational datasets in sport. Every race weekend produces 50,000+ data points per car. The client needs a platform that joins 74 years of history with live race-day intelligence to make faster, better-informed decisions.

The four dimensions of the challenge:

### VOLUME

589K lap-time records
14 tables · 1950–2024

### VELOCITY

Pit decisions in seconds
Live telemetry

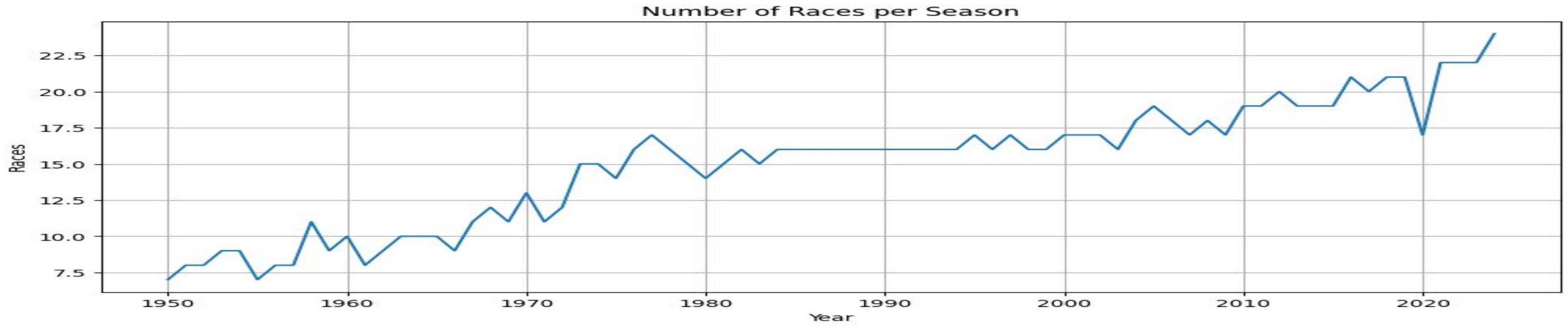
### VARIETY

Results, quali, pit stops
standings, laps

### VERACITY

Missing data · Status inconsistencies
Era schema drift

*Calendar growth: races per season 1950–2024 (Source: Notebook 2)*

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TECHNOLOGY CHOICE

# Why Apache Spark?

- **Processes hundreds of thousands of lap records today scales to millions as live telemetry expands**

  A single race weekend produces 50,000+ data points per car. Spark distributes this across cores automatically, no manual optimisation required.

- **One platform: history, ML, and live data**

  Spark handles batch analytics, machine learning pipelines, and live streaming in a single system. No tool-switching mid-race.

- **Scales horizontally as data grows**

  As telemetry sensors, weather feeds, and tyre data are added the platform scales out,  not up. No re-architecture required.

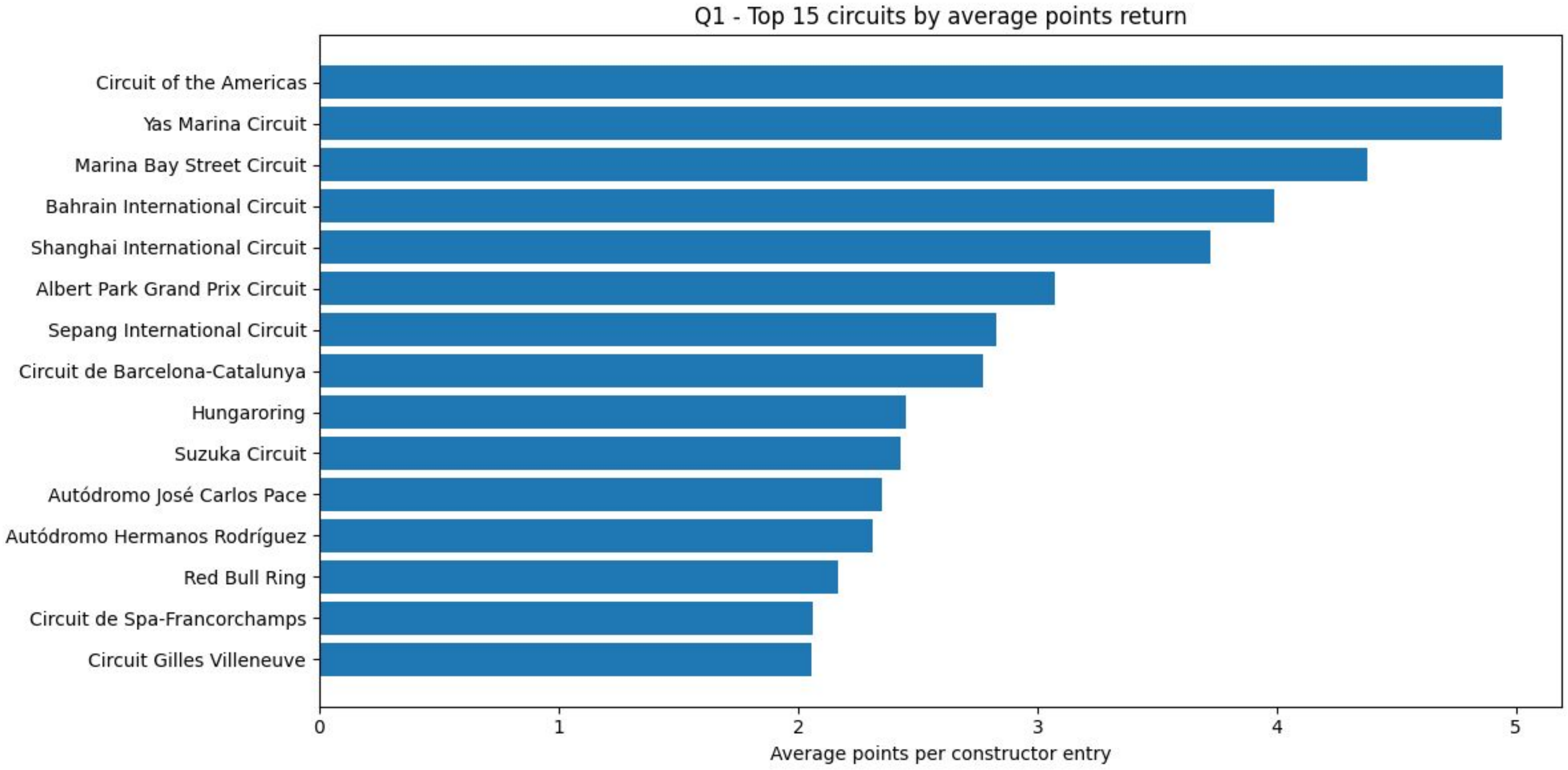
- **Fault-tolerant by design**

  If a node fails mid-calculation Spark re-runs only the affected partition automatically. No data loss, no manual recovery.

*Bottom line: Spark is the only open-source engine that does all of this without leaving the platform.*

## STRATEGIC INSIGHT 1 / 2

*Circuit Profitability — avg points return per constructor entry by circuit*



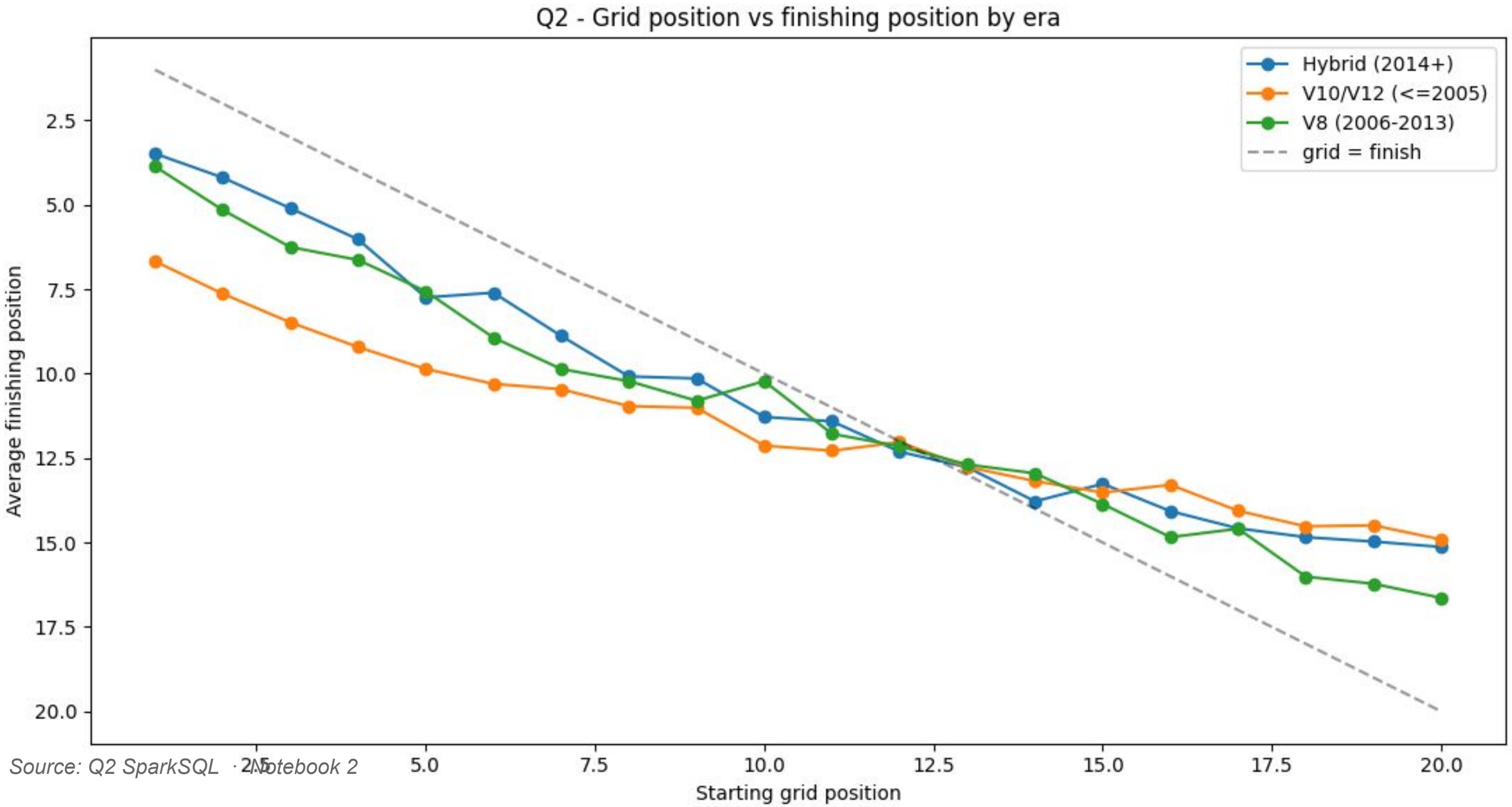

*Source: Q1 SparkSQL · Notebook 2*

### Key Findings

- COTA & Yas Marina top the table (~5.0 pts/entry). Both were added after 2010 when the points system tripled from 10 to 25 pts per win, making their average structurally higher.
- Marina Bay (Singapore) ranks 3rd, a street circuit that rewards reliability over outright pace
- All top-5 circuits entered the calendar after 2004. Spa, Silverstone and Monza carry 50+ years of 8 to 10 pt race data that pulls their all-time average down.
- Recommendation: treat the ranking as a points era effect, not circuit quality. The real strategic metric is average points at circuits still on the current calendar under the 25 point system.

**STRATEGIC INSIGHT 2 / 2**

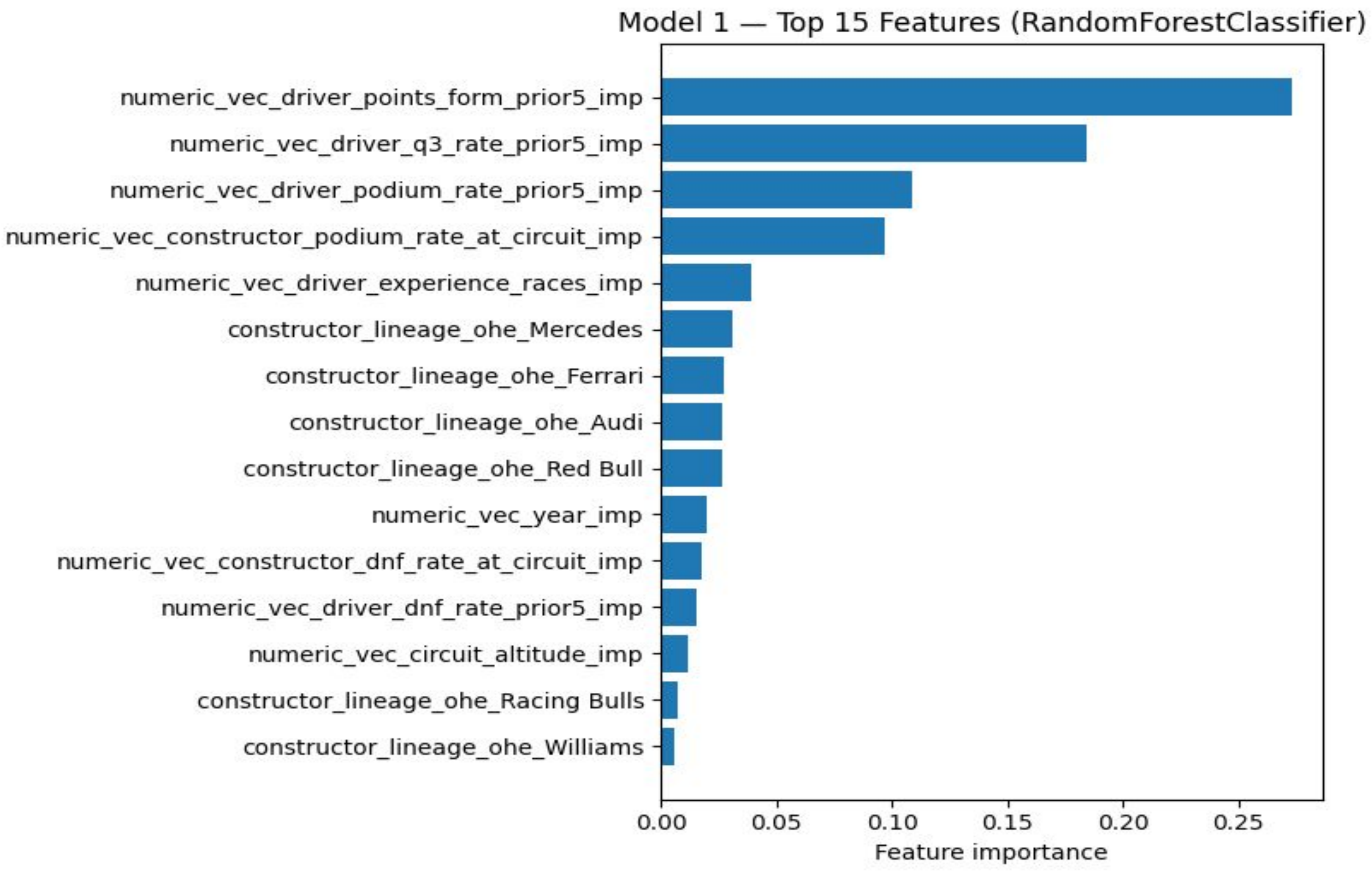
*Does Qualifying Actually Matter? — grid position vs finishing position by era*




*Source: Q2 SparkSQL · Notebook 2*

## Key Findings

- Pole position strongly correlates with top-3 finishes in the hybrid era, the chart shows grid 1 delivers the lowest avg finish position across all eras
- Hybrid era pole converts best, grid 1 averages ~3rd place vs ~5th (V8 era) and ~7th (V10/V12 era); the qualifying premium has grown each era
- Circuits with the highest grid-finish correlation (lowest overtaking) show near-linear conversion, starting mid-pack offers minimal recovery; ovals/open circuits show the opposite
- Implication: weight Saturday qualifying more heavily in the development budget



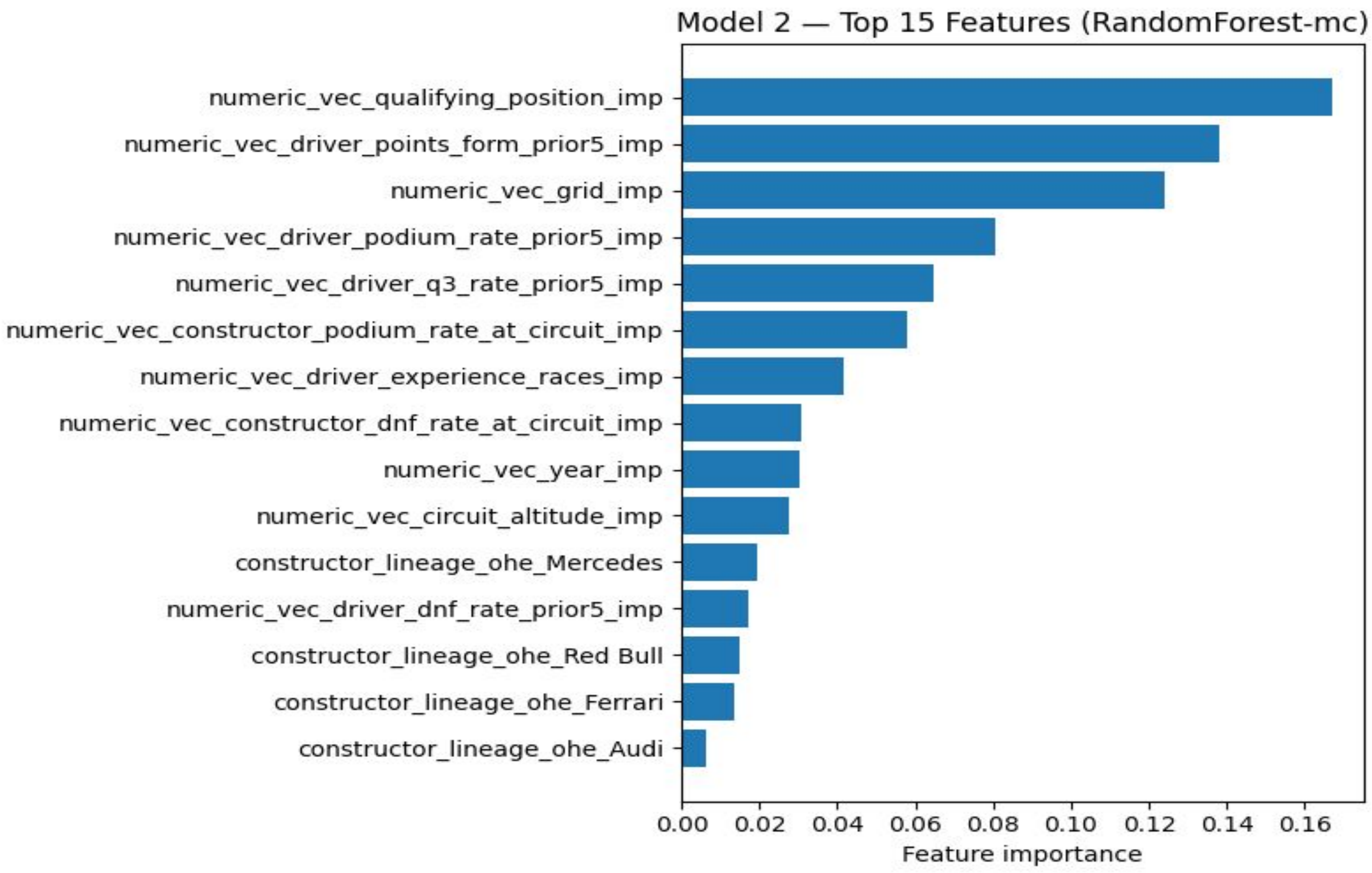

PREDICTION TOOL 1 / 2

# Will We Make Q3?

## Qualifying Predictor

- Predicts whether a driver will reach Q3 before the weekend begins
- 88% ROC-AUC · 78% accuracy · PR-AUC = 0.878 · 5-fold CV
- Winner: RandomForestClassifier · 80/20 held-out test set (823 rows)
- Outputs a probability score, the team sets the risk threshold
- Benchmarked against GBT, MLP, and LogReg RF wins on both CV-AUC and test-AUC

*Source: Notebook 3 · RandomForestClassifier + LogisticRegression baseline*




PREDICTION TOOL 2 / 2

# What Will Our Race Outcome Be?

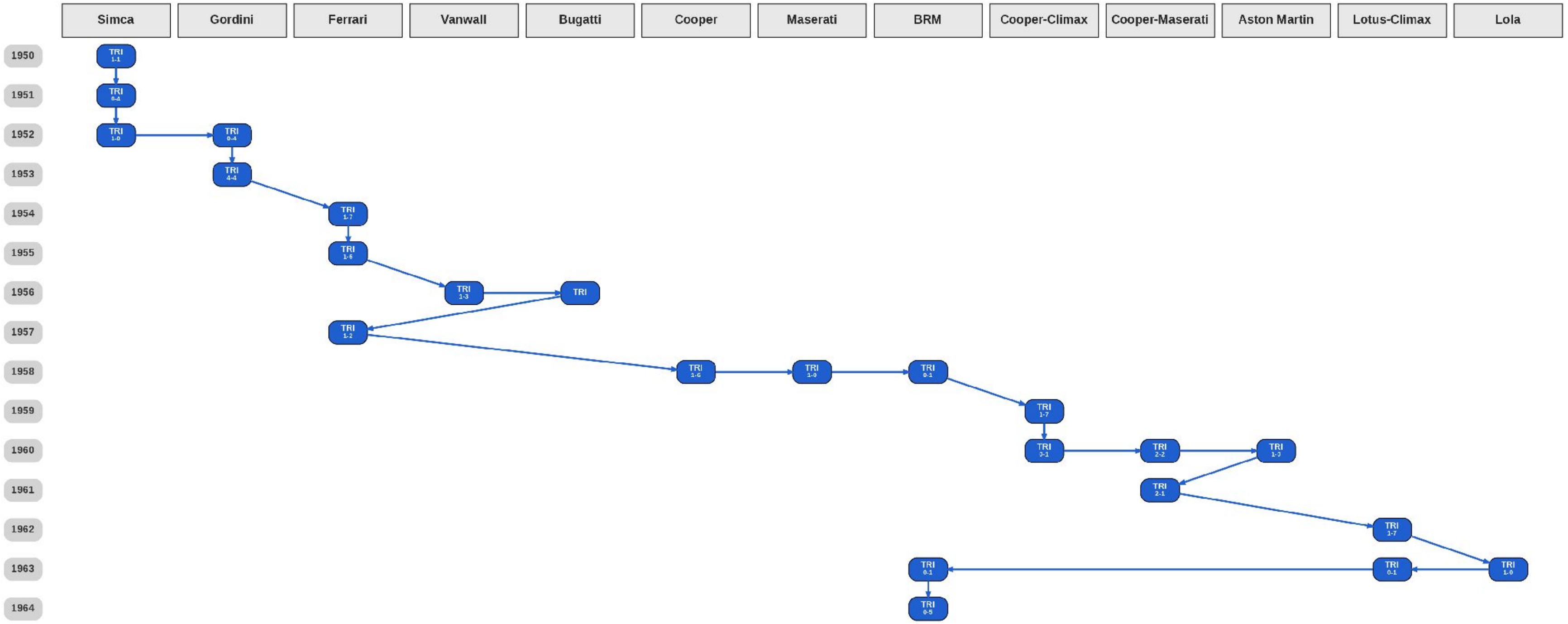
## Race Outcome Predictor

- 5-class prediction: Win · Podium · Points · Finisher-NP · DNF
- Weighted F1 = 0.51 · Accuracy = 54% · Baseline (majority class) = 20% F1
- Qualifying position is the single strongest predictor, confirms Saturday's priority
- Constructor circuit history is second, prior podium rate at a venue is a genuine edge
- Provides probability distribution across all 5 outcomes, not just a binary flag

*Source: Notebook 3 · RandomForest-mc · TrainValidationSplit + held-out test (80/20)*

## GRAPH ANALYTICS

*Why Does Maurice Trintignant Top the PageRank? · 861-driver teammate network · 16,042 edges*




*Maurice Trintignant · 13 constructors · 1950–1964 (Notebook 2 · describe_cluster())*

### PageRank 5.70

*Highest of all 861 drivers*

- 13 constructors over 14 seasons, each team adds entirely new teammates to his edge list
- Multi-team years (1956, 1960, 1963) multiply his connections far beyond single-team drivers
- 2nd: Stirling Moss 5.49 · 3rd: Jack Brabham 5.01, all era-bridging drivers
- Constructor lineage DAG: 32 nodes · 22 edges, Aston Martin highest dynasty PageRank

*Source: Notebook 4 · GraphFrames PageRank + LPA + BFS · Notebook 2 · describe_cluster()*

## LIVE RACE DASHBOARD · BONUS

*Spark Structured Streaming — Abu Dhabi Grand Prix 2022*

**LIVE LEADERBOARD | Abu Dhabi 2022 | Lap 1**

| P | COD | DRIVER | LAP TIME | GAP / ΔPOS |
|---|---|---|---|---|
| 1 | VER | Max Verstappen | 1:32.198 | LEADER |
| 2 | PER | Sergio Pérez | 1:33.251 | +1.053s |
| 3 | LEC | Charles Leclerc | 1:34.192 | +1.994s |
| 4 | HAM | Lewis Hamilton | 1:34.864 | +2.666s ▲1 |
| 5 | SAI | Carlos Sainz | 1:35.436 | +3.238s ▼1 |
| 6 | NOR | Lando Norris | 1:35.908 | +3.710s ▲1 |
| 7 | RUS | George Russell | 1:36.506 | +4.308s ▼1 |
| 8 | OCO | Esteban Ocon | 1:36.964 | +4.766s |

### How it works

- readStream with maxFilesPerTrigger=1, lap files dropped every 2 s
- withWatermark (10 s) handles late-arriving telemetry without blocking
- foreachBatch delivers each micro-batch to the live leaderboard handler
- Position changes (▲▼) computed in real time from prior-lap window
- Audit log written to Parquet in append mode, immutable race record
- Validated on Abu Dhabi 2022 · fastest lap 89,968 ms · 20 drivers

*Source: Notebook 5 · Spark Structured Streaming · actual notebook output shown*

# Thank You!

Any questions?

*F1 Race Strategy Intelligence Platform · NOVA IMS · Big Data Analytics 2026*